Reg. No. | | | | | | | | | |

Question Paper Code : 16384

B.E. / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2020

Third Semester

B.E. – Computer Science and Engineering

19CS391 – PROGRAMMING WITH JAVA

(Regulations: Mepco – R2019)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL, CO

PART A – (10 × 2 = 20 Marks)

R, CO1 1. List the access specifiers used in java programming.

A,CO1 2. What will be the output of the following program? Justify your answer.

```
public class Test {
public static void main (String [] args) {
   int count = 1;
   while (count <= 15) {
   System.out.println(count % 2 == 1? "***" : "+++++");
   ++count;
      }     // end while
   }      // end main
 }
```

A) 8 times *** B) 15 times ***

C) 8 times *** and 7 times +++++ D) 15 times +++++

R, CO2 3. Define autoboxing with an example.

E,CO2 4. An interface with no fields or methods is known as _____ interface. Justify your case.

A) Runnable B) Abstract C) Marker D) Functional

R,CO3 5. Consider a C function doubleValue (int x) that prints twice the value of x. Write a Java class that invokes this method using native method.

L,CO3 6. Which of these is a correct way of defining generic method “name”? Justify your answer.

A) <T1, T2, …, Tn> name {T1, T2, …, Tn} {/* … */}

B) <T1, T2, …, Tn> name (T1, T2, …, Tn) {/* … */}

C) public <T1, T2, …, Tn> name<T1, T2, …, Tn> {/* … */}

D) class <T1, T2, …, Tn> name [T1, T2, …, Tn] {/* … */}

U,CO5 7. How to create and position the frame using swing? Give an example.

R,CO5 8. The _____ listener interface is associated with the selection of check box AWT control. Justify your answer.

A) ActionListener B) ItemListener

C) KeyListener D) MouseListener

R,CO6 9. Recall the thread properties.

L,CO6 10. What will happen if two threads of the same priority are called to be processed simultaneously? Justify your answer.

A) Both of them will be executed simultaneously

B) None of them will be executed

C) It is dependent on the operating system

D) Anyone will be executed first

PART B – (5 × 16 = 80 Marks)

A,CO1 11. a) i. Write a java program to calculate the total and average of ten students mark details using array, strings and static members. (10 Marks)

U,CO1 11. a) ii. Give a brief account on Enumeration class. (6 Marks)

OR

A,CO1 11. b) i. Write a Java program that implements bubble sort algorithm for sorting a list of names in ascending order. (10 Marks)

U,CO1 11. b) ii. How to use String Object as expression in switch statement? Explain with example. (6 Marks)

A,CO2 12. a) i. Construct a Message abstract class which contains playMessage( ) abstract method. Write different sub-classes like TextMessage, VoiceMessage and FaxMessage classes for implementing the playMessage( ) method. Show how to invoke this method using polymorphism. (10 Marks)

U,CO2 12. a) ii. Explain the steps involved in exception handling in detail. (6 Marks)

OR

A,CO2	12. b) i.	Using functional interface, define a functional interface that computes the factorial of a given number. Write suitable Java program to implement the same.	(10 Marks)
U,CO2	12. b) ii.	Explain the steps involved in JAR file creation.	(6 Marks)
A,CO3	13. a) i.	Write a Java program for the following: A) Create a doubly linked list of elements B) Delete a given element from the above list C) Display the contents of the list after deletion (Note: The method should be able to accept both integer arrays and string arrays)	(8 Marks)
U,CO3	13. a) ii.	Define object serialization and explain the steps involved for loading and saving serialized objects.	(8 Marks)
		OR	
A,CO3	13. b) i.	Write suitable program that implements HashMap and TreeMap for storing employee details.	(8 Marks)
U,CO3	13. b) ii.	Explain in detail generic programming with an example.	(8 Marks)
U,CO5	14. a) i.	Explain about applet life cycle and how applets are prepared and executed.	(8 Marks)
A,CO5	14. a) ii.	Develop a Java swing program that receives an integer in one text field, and computes its factorial value and returns it in another text field, when the button named “Compute” is clicked.	(8 Marks)
		OR	
U,CO5	14. b) i.	Explain any four JavaFX components in detail.	(8 Marks)
A,CO5	14. b) ii.	Develop an editor like Notepad++ using swings menus and dialog boxes.	(8 Marks)
A,CO6	15. a) i.	Write a java program that implements a multi-thread application that has three threads. First thread generates random integer every one second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number.	(10 Marks)

U,CO6	15. a) ii.	Explain briefly the inter thread communication process with an example.	(6 Marks)

OR

A,CO6	15. b) i.	Write a java program to show multithreaded producer and consumer application.	(10 Marks)
U,CO6	15. b) ii.	Show the life cycle of multithreading, with a neat diagram.	(6 Marks)